

POLYNOMIAL REGRESSION

MODEL SELECTION & VALIDATION

Phase 1: degree selection | Phase 2: regularization and split comparison

Submission report · 8 October 2026 · Prepared from supplied experimental output
Machine Learning Assignment - 1 | Roll No:IMT2024091 Name : Puttaraja beedinalmath
K Git hub repo link: https://github.com/puttasworkspace/ML-PROJECT_LINEAR-REGRESSION.git

Executive Summary

This report documents a two-phase polynomial-regression experiment. Phase 1 compares polynomial degrees 1–7 for Ridge, Lasso, and Elastic Net, selects degree by validation RMSE, then reports the selected models after refitting on the combined training and validation data. Phase 2 expands the degree search to 1–20, tunes regularization, and compares 70/30, 75/25, 80/20, 85/15, and 90/10 train–test splits using cross-validation summaries and held-out test metrics.

Headline results

Phase 1: Lasso, degree 5, is the strongest overall candidate (validation RMSE 0.5864; after refit, test RMSE 0.5682, MAE 0.4592, R^2 0.9629). Phase 2: the requested 80/20 Ridge candidate is degree 12 with $\alpha = 3.1623$ (CV RMSE 0.5119; test RMSE 0.4947, MAE 0.3838, R^2 0.9932). The 90/10 Ridge degree-12 result has a lower test RMSE (0.4788), but it is measured on a different, smaller holdout.

1. Phase 1 — Polynomial Degree and Model Selection

The first phase evaluates the effect of polynomial degree while comparing three regularized linear estimators. The supplied results report training, validation, and internal-test RMSE and R^2 for degrees 1–7. The selection rule is the lowest validation RMSE within each model family; the final comparison then refits each selected configuration on training plus validation data and scores it on the internal test set.

Validation RMSE across the degree search

Degree	Ridge	Lasso	Elastic Net
1	3.2424	3.2399	3.2421
2	1.9325	1.9315	1.9316
3	1.0944	1.0818	1.0829
4	0.8997	0.8188	0.8297
5	0.7720	0.5864	0.5909
6	0.7229	0.5952	0.6022
7	0.7845	0.5937	0.5982

Selected configurations and final refit results

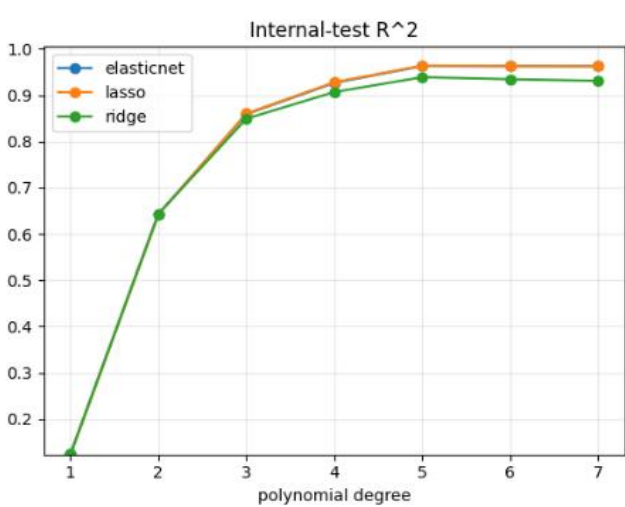
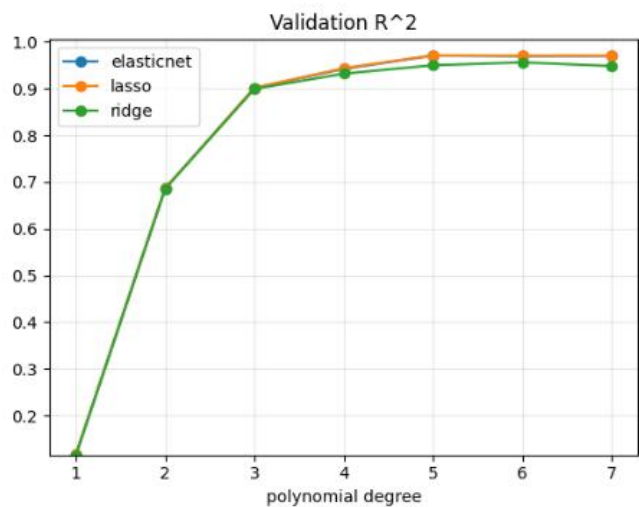
Model	Degree	Val. RMSE	Val. R ²	Test RMSE	Test MAE	Test R ²
Ridge	6	0.7229	0.9560	0.7583	0.5929	0.9340
Lasso	5	0.5864	0.9711	0.5682	0.4592	0.9629
Elastic Net	5	0.5909	0.9706	0.5761	0.4651	0.9619

Interpretation. Increasing degree from 1 to 5 sharply reduces validation error, showing that a straight-line or low-order relationship is not sufficient for this task. Lasso reaches its minimum at degree 5 (0.5864); Elastic Net is close (0.5909). Degrees 6–7 do not improve either model, so higher complexity provides no validation benefit. Ridge improves through degree 6 (0.7229) but worsens at degree 7 (0.7845), consistent with the practical risk that extra polynomial terms can raise variance without improving generalization.

Phase 1 conclusion

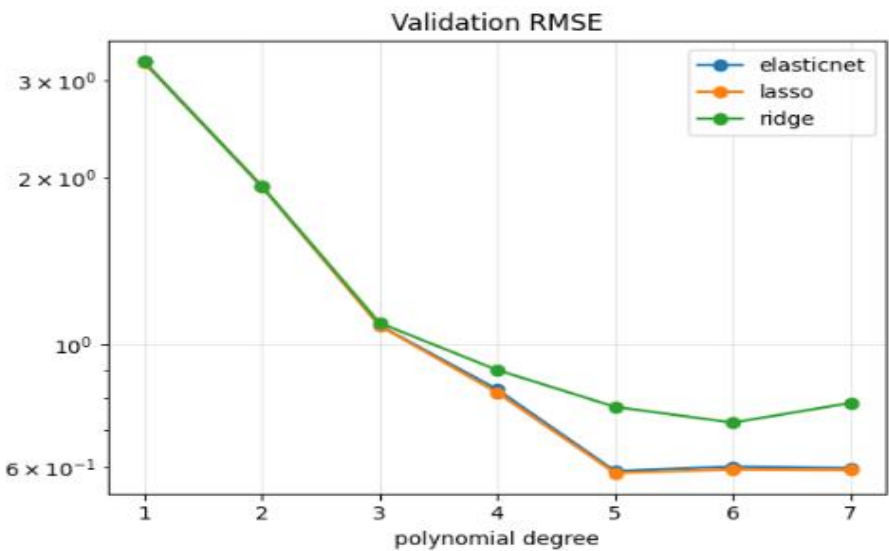
If one model must be carried forward from Phase 1, Lasso degree 5 is the evidence-based choice: it has the lowest validation RMSE and, after refitting, the lowest reported test RMSE and MAE and the highest test R² among the three selected models.

The Lasso–Elastic Net differences are modest, so the results support a clear ranking but not a claim that their predictive quality is dramatically different. The degree-5 choice also favors a simpler polynomial than degrees 6 or 7 without sacrificing observed validation accuracy.



[validation R2 score vs the polynomial degree](#)

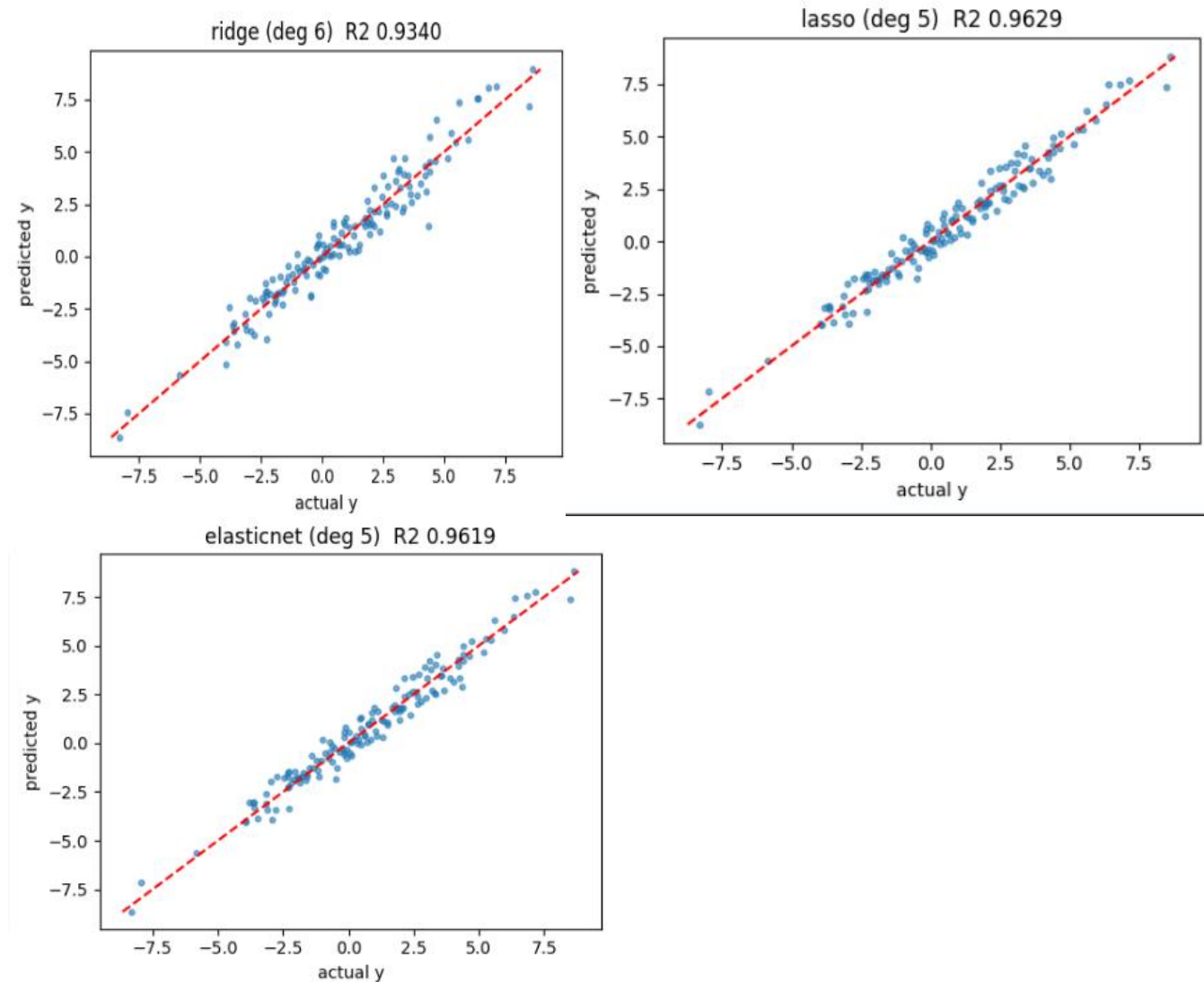
[Internal test R2 vs the polynomial degree](#)



[Validation RMSE vs the polynomial degree](#)

Comparing all the models of across L1,L2, and elastic nets

Predicted Y vs the Actual Y data



2. Phase 2 — Regularization and Split Comparison

The second phase broadens the polynomial-degree search to 1–20 and evaluates Ridge, Lasso, and Elastic Net under five train–test proportions. The result tables vary the polynomial degree and regularization strength (α); Elastic Net additionally varies $l1_ratio$ (reported values include 0.2, 0.5, and 0.8). Candidate rankings report cross-validation RMSE and R^2 means and standard deviations, alongside training metrics. The summary tables give one tuned configuration per estimator and split, followed by held-out test metrics.

Best cross-validation RMSE candidate by split

Split	Lowest-CV-RMSE configuration*	CV RMSE	Test RMSE	Test MAE	Test R^2
90/10	Ridge · d12 · $\alpha=1$	0.4864	0.4788	0.3809	0.9923
85/15	Ridge · d12 · $\alpha=1$	0.4943	0.5173	0.4099	0.9911
80/20	Elastic Net · d14 · $\alpha=0.00215$ · $l1=0.2$	0.5033	0.4951	0.3827	0.9931
75/25	Ridge · d10 · $\alpha=1$	0.5027	0.5053	0.3961	0.9934
70/30	Ridge · d10 · $\alpha=1$	0.5212	0.4894	0.3843	0.9943

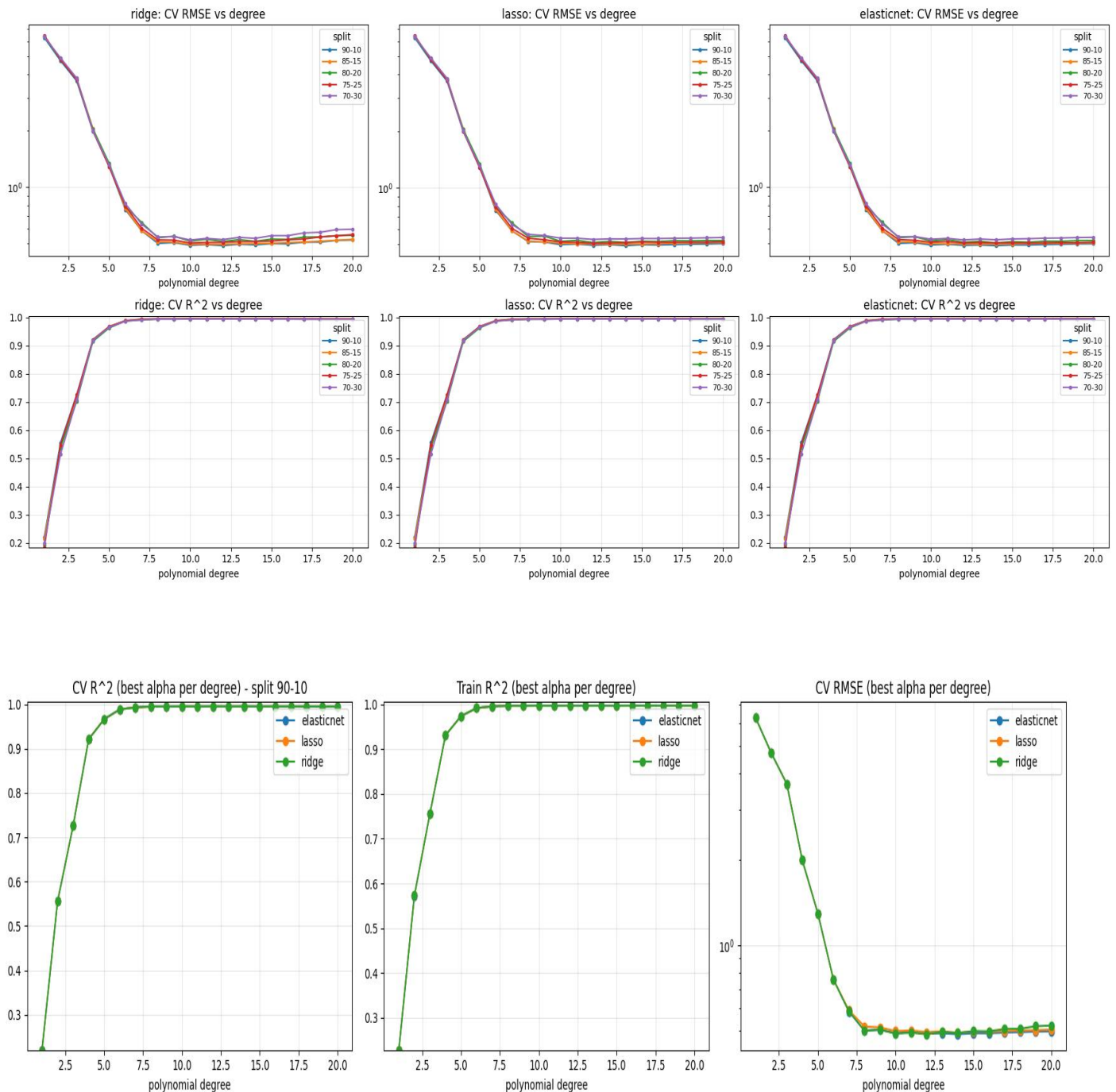
*Among the three model-level configurations in the supplied final comparison for each split. At 90/10, Ridge and Elastic Net are effectively tied on CV RMSE (0.486396 vs. 0.486431); the displayed rounding should not be read as a meaningful separation.

What the split comparison shows

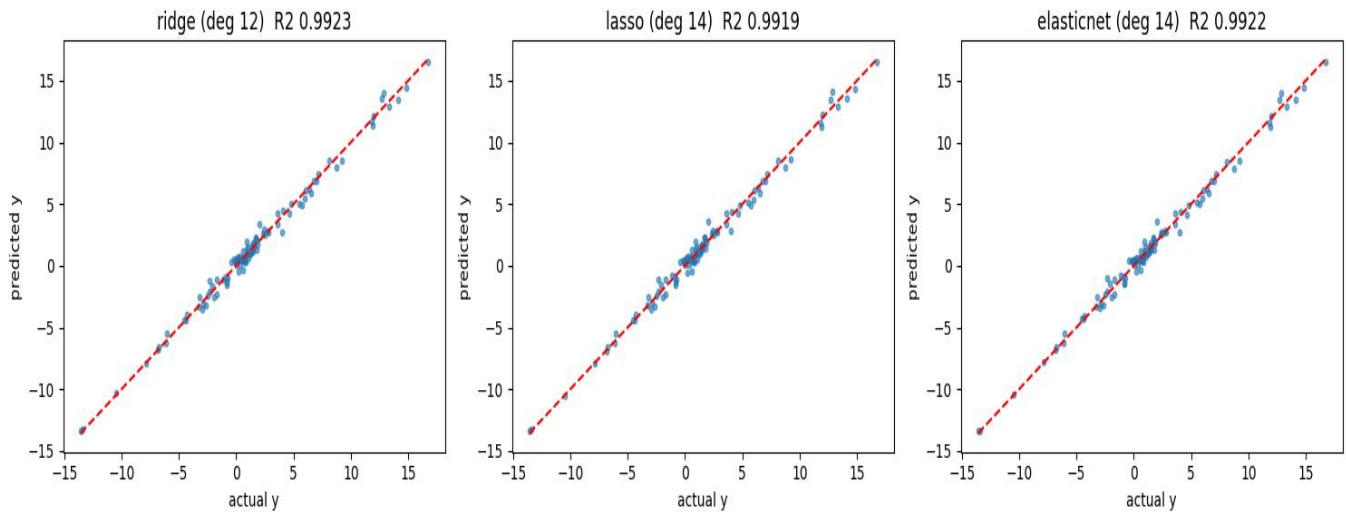
- Ridge selects degree 12 for the 90/10, 85/15, and 80/20 settings, then degree 10 for 75/25 and 70/30. This repeated degree-12 choice across the larger-training splits is useful evidence that the Ridge degree selection is not an isolated 80/20 result.
- The lowest-CV-RMSE family is Ridge for 90/10, 85/15, 75/25, and 70/30, but Elastic Net for 80/20. Thus no single family wins every split by cross-validation.
- Test RMSE does not move monotonically with the split ratio. Each ratio evaluates on a different held-out sample; differences can reflect sample composition as well as model fit. R^2 likewise depends on the target variation within that particular holdout.

Evaluation principle

Use cross-validation for tuning and model selection, then treat the test set as a final estimate—not as a second tuning set. Compare test scores across split ratios cautiously unless the same observations and a paired evaluation design are used.



Above are the plots of the CV R², Train R² and the CV RMSE across the different polynomial degree



All the best models across different degrees

3. Why Ridge Degree 12 at 80/20 Is Defensible

The 80/20 Ridge configuration reported in the supplied summary is degree 12 with $\alpha = 3.1623$. Within the Ridge family, it is the configuration selected for that split; its cross-validation RMSE is 0.5119 and its held-out test RMSE is 0.4947. The table below places it beside the 90/10 Ridge result and the other 80/20 families so the rationale is explicit.

Configuration	Degree	Penalty	CV RMSE	Test RMSE	Test MAE	Test R ²
90/10 Ridge	12	1.000	0.4864	0.4788	0.3809	0.99235
80/20 Ridge	12	3.162	0.5119	0.4947	0.3838	0.99315
80/20 Lasso	12	0.000464	0.5112	0.5023	0.3925	0.99294
80/20 Elastic Net	14	0.002154; l1=0.2	0.5033	0.4951	0.3827	0.99314

Rationale for the 80/20 choice

- A 20% holdout gives the final evaluation twice the test fraction of 90/10 and therefore more held-out cases, assuming a representative split. The trade-off is that 80/20 uses fewer cases for fitting than 90/10. The larger holdout is a defensible evaluation-design preference, not a guarantee of a lower error.
- Degree 12 is consistent within Ridge across 90/10, 85/15, and 80/20. This stability supports choosing degree 12 for the 80/20 Ridge run rather than treating it as a one-off optimum.
- The 90/10 Ridge test RMSE is lower by 0.0159 (about 3.3%) than the 80/20 Ridge test RMSE. Because the test observations differ, that gap does not prove that 90/10 is the better generalizing model. The 80/20 Ridge test R² is slightly higher, but that metric also is not a paired comparison across different test samples.

Important qualification

What the numbers do—and do not—support

Across all three 80/20 families, Elastic Net has the lowest CV RMSE (0.5033); Ridge is higher at 0.5119. On the held-out test, Ridge has the lowest RMSE (0.494731) versus Elastic Net (0.495140), an advantage of only 0.000408 (about 0.08%); Elastic Net has the lower MAE by 0.001052, and the test R² values are practically indistinguishable. Therefore, Ridge degree 12 is defensible as the chosen 80/20 Ridge candidate, but the supplied results do not establish it as the unambiguous overall 80/20 winner. If the required rule is strictly the lowest cross-validation RMSE across families, the 80/20 choice should be Elastic Net degree 14.

4. Interpretation, Limitations, and Conclusion

Techniques used

- Polynomial feature expansion: candidate degree controls the order of nonlinear terms available to the linear estimator. Increasing degree can capture curvature but increases model flexibility and the risk of overfitting.
- Regularization: Ridge applies an L2 penalty to shrink coefficient magnitudes; Lasso applies an L1 penalty and can set some coefficients to zero; Elastic Net combines the two, with `l1_ratio` setting the mixture.
- Validation and cross-validation: validation/CV RMSE is used to compare degrees and penalty settings, while R^2 provides a scale-free measure of explained variance. The final test metrics summarize performance on observations not used to tune the selected configuration, as described in the supplied Phase 1 results.

Metric interpretation

RMSE and MAE are expressed in the response variable's units; lower is better. RMSE penalizes larger errors more heavily than MAE. R^2 is a proportion-like measure of fit relative to predicting the target mean; higher is better, though its value can change with the composition and spread of each test set. The target units and scale were not included in the supplied material, so errors are reported without a unit label.

Limitations and reproducibility

The attachment contains result tables but not the dataset or executable code. Consequently, the report cannot verify feature/target definitions, preprocessing or scaling, missing-value handling, split random seed, cross-validation fold count and strategy, leakage controls, or residual diagnostics. No confidence intervals or repeated-split distributions are supplied. These omissions limit claims about reproducibility and statistical significance; the very small 80/20 Ridge-versus-Elastic-Net test differences should be interpreted as near-ties rather than decisive evidence.

Conclusion

Phase 1 supports Lasso degree 5 as the best overall candidate among the tested models: it minimizes validation RMSE and also leads the refitted test comparison. Phase 2 supports Ridge degree 12 as a stable choice within the Ridge family for an 80/20 split, with strong held-out fit (test R^2 0.9932). The lower 90/10 Ridge RMSE is not a direct head-to-head result because the holdout differs. The final report should therefore state the split and selection rule clearly: retain Ridge degree 12 if the objective is the chosen 80/20 Ridge model and a larger holdout; report Elastic Net degree 14 if the objective is strictly the lowest 80/20 cross-validation RMSE across all families.